**Project Design Phase-I**
**Proposed Solution Template**

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| Date | 26 February 2026 |
| Team ID | PNT2022TMIDxxxxxx |
| Project Name | An Exploration of Electricity Consumption Patterns |
| Maximum Marks | 2 Marks |

**Proposed Solution Template:**

| S.No. | Parameter | Description |
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| 1. | Problem Statement (Problem to be solved) | Consumers lack clear visibility into their electricity consumption patterns, leading to energy wastage, high electricity bills, and inefficient usage of electrical appliances. The absence of real-time monitoring and awareness makes it difficult for users to identify peak usage periods and adopt energy-saving practices. |
| 2. | Idea / Solution description | The proposed solution is a smart electricity consumption analysis system that collects and analyzes usage data to identify patterns and inefficiencies. The system provides users with easy-to-understand insights, usage trends, and recommendations to reduce unnecessary consumption. It helps users make informed decisions about energy usage, promotes efficient appliance usage, and supports sustainable energy management. |
| 3. | Novelty / Uniqueness | Unlike traditional electricity billing systems that provide only monthly consumption data, this solution focuses on pattern-based analysis and user awareness. It offers actionable insights rather than raw data, enabling users to understand when, where, and how electricity is consumed. The solution emphasizes behavioral change along with technology-driven analysis. |
| 4. | Social Impact / Customer Satisfaction | The solution promotes energy conservation, reduces electricity expenses for consumers, and contributes to environmental sustainability by lowering carbon emissions. Increased awareness |

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| | | and control over energy usage lead to higher customer satisfaction, improved quality of life, and responsible energy consumption within communities. |
| 5. | Business Model (Revenue Model) | The solution can operate on a subscription-based model for advanced analytics and insights. Additional revenue can be generated through partnerships with utility providers, smart appliance manufacturers, and energy consulting services. Basic features can be offered free to encourage adoption, with premium features available at a nominal cost. |
| 6. | Scalability of the Solution | The solution is highly scalable as it can be implemented across residential, commercial, and industrial sectors. With cloud-based infrastructure and data analytics, it can support a growing number of users without significant changes to the core system. The solution can also be expanded to integrate renewable energy monitoring and smart grid systems in the future. |